

Hands-on Lab: Create Tables using SQL Scripts and Load Data into Tables in MySQL using phpMyAdmin



Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Software Used in this Lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Database Used in this Lab

The database used in this lab is an internal database. You will be working on a sample HR database. This HR database schema consists of 5 tables called **EMPLOYEES**, **JOB_HISTORY**, **JOBS**, **DEPARTMENTS** and **LOCATIONS**. Each table has a few rows of sample data. The following diagram shows the tables for the HR database:

SAMPLE HR DATABASE TABLES										
EMPLOYEES										
EMP_ID	LAST_NAME	FIRST_NAME	DOB	D_DATE	SEX	ADDRESS	JOB_ID	SALARY	MANAGER_ID	DEPT_ID
E1001	John	Thomas	123456	1976-01-09	M	5011 Hwy OakfordA	100	100000	30001	2
E1002	Alice	James	123457	1972-07-31	F	807 Berry Ln,SpicA	200	80000	30002	5
E1003	Steve	Wells	123458	1980-08-10	M	291 Spring, GensA	300	50000	30003	7
JOB_HISTORY										
EMPL_ID	START_DATE	JOB_ID	DEPT_ID							
E1001	2000-01-01	100	2							
E1002	2010-08-16	200	5							
E1003	2016-08-10	300	5							
JOBS										
JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY							
100	Sr Architect	40000	100000							
200	Sr Software Developer	60000	80000							
300	Jr Software Developer	40000	60000							
DEPARTMENTS										
DEPT_ID	DEPT_NAME	MANAGER_ID	LOC_ID							
2	Architect Group	30001	10001							
5	Software Development	30002	10002							
7	Design Team	30003	10003							
LOCATIONS										
LOC_ID	LOC_NAME	LOC_ADDRESS	LOC_CITY	LOC_STATE	LOC_COUNTRY	LOC_POSTAL_CODE	LOC_PHONE	LOC_FAX	LOC_EMAIL	
10001	10001	10001	10001	10001	10001	10001	10001	10001	10001	
10002	10002	10002	10002	10002	10002	10002	10002	10002	10002	
10003	10003	10003	10003	10003	10003	10003	10003	10003	10003	

Objectives

After completing this lab, you will be able to use phpMyAdmin with MySQL to:

- Create a database.
- Create tables using SQL scripts
- Load data into tables

Exercise

In this exercise through different tasks, you will learn how to create tables and load data in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

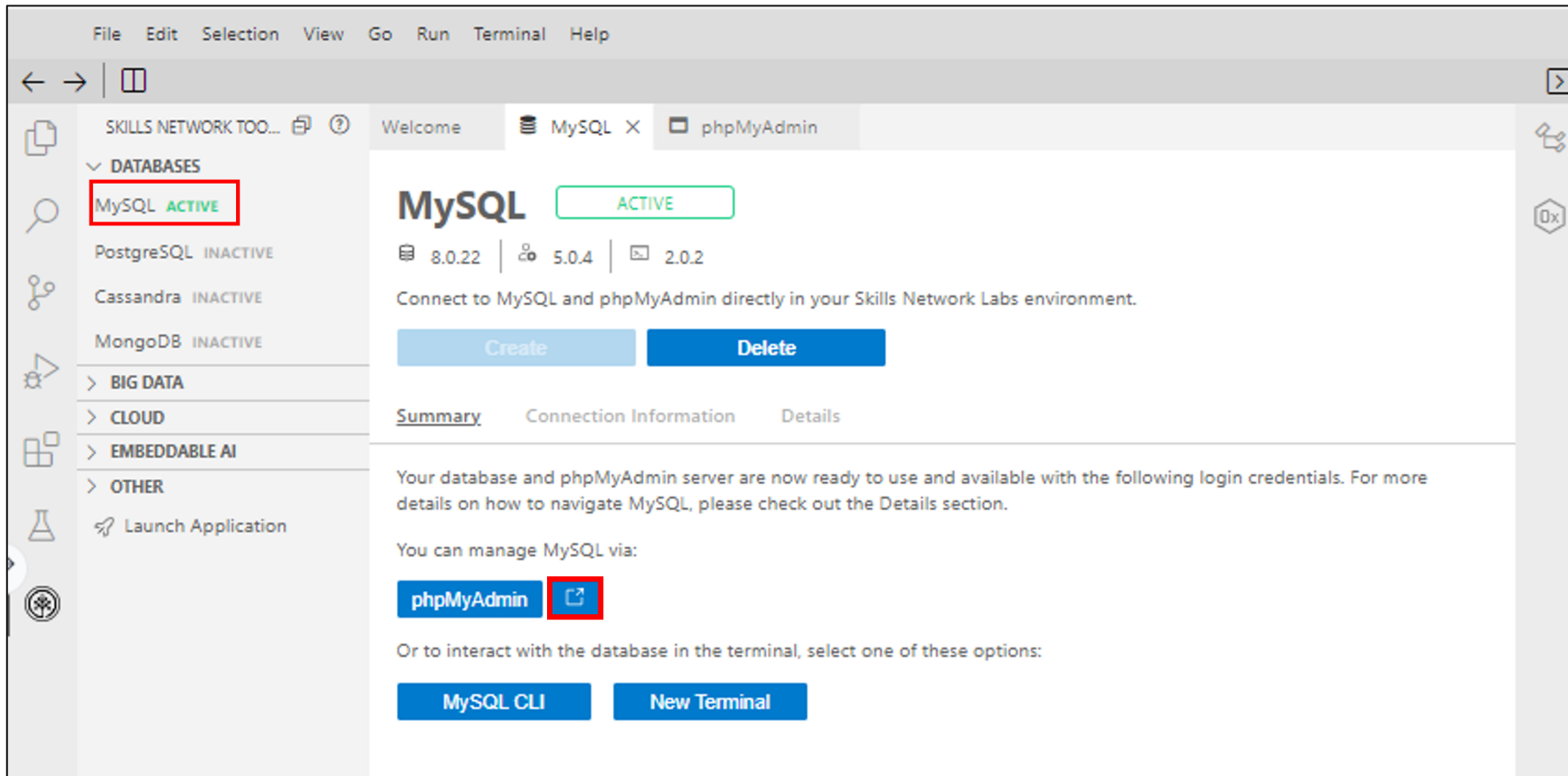
Task A: Create a database

1. Click on **Skills Network Toolbox**. Under **Database** section, click **MySQL**.

To start the MySQL, click **Create**.

The screenshot shows the Skills Network Labs IDE interface. The menu on the left is expanded to show the 'DATABASES' section. The 'MySQL INACTIVE' entry is highlighted with a red box and a red circle with the number 1. The 'Create' button is highlighted with a red box and a red circle with the number 2. The main panel displays the MySQL database status as 'INACTIVE' and provides options to 'Create' or 'Delete' the database.

2. Once MySQL has started, click on **phpMyAdmin** button to open **phpMyAdmin** in the same window.



File Edit Selection View Go Run Terminal Help

SKILLS NETWORK TOO... Welcome MySQL X phpMyAdmin

DATABASES

- MySQL **ACTIVE**
- PostgreSQL INACTIVE
- Cassandra INACTIVE
- MongoDB INACTIVE

BIG DATA

CLOUD

EMBEDDABLE AI

OTHER

Launch Application

MySQL

ACTIVE

8.0.22 | 5.0.4 | 2.0.2

Connect to MySQL and phpMyAdmin directly in your Skills Network Labs environment.

Create Delete

Summary Connection Information Details

Your database and phpMyAdmin server are now ready to use and available with the following login credentials. For more details on how to navigate MySQL, please check out the Details section.

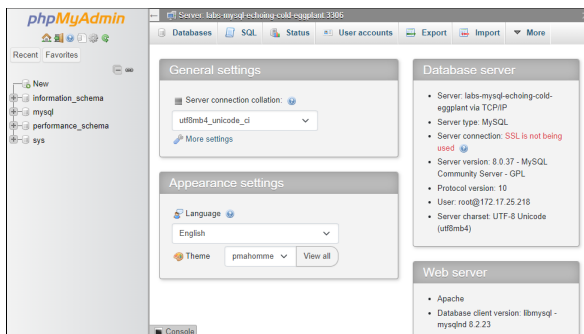
You can manage MySQL via:

phpMyAdmin

Or to interact with the database in the terminal, select one of these options:

MySQL CLI New Terminal

3. You will see the phpMyAdmin GUI tool.



phpMyAdmin

Recent Favorites

New

- information_schema
- mysql
- performance_schema
- sys

General settings

Server connection collation: utf8mb4_unicode_ci

More settings

Appearance settings

Language: English

Theme: pinahomme View all

Database server

- Server: labs-mysql-echoing-cold-eggplant.3386
- Server type: MySQL
- Server connection: SSL is not being used
- Server version: 8.0.37 - MySQL Community Server - GPL
- Protocol version: 10
- User: root@172.17.25.219
- Server charset: UTF-8 Unicode (utf8mb4)

Web server

- Apache
- Database client version: libmysql - mysqlnd 8.2.23

4. In the tree-view, click **New** to create a new empty database. Then enter **HR** as the name of the database and click **Create**.

The encoding will be left as **utf8mb4_0900_ai_ci**. UTF-8 is the most commonly used character encoding for content or data.

Proceed to Task B.

phpMyAdmin

Server: labs-mysql-echoing-cold-eggplant3306

Databases SQL Status User accounts Export Import Settings

Databases

Create database

2 HR 3 utf8mb4_0900_ai_ci 4 Create

☐ Check all Search

	Database	Collation	Primary replication	Action
☐	information_schema	utf8mb3_general_ci	✓ Replicated	Check privileges
☐	mysql	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges
☐	performance_schema	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges
☐	sys	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges

Total: 4

Exercise 1: Create tables using SQL scripts

In this exercise, you will learn how to execute a script containing the CREATE TABLE commands for all the tables rather than create each table manually by typing the DDL commands in the SQL editor.

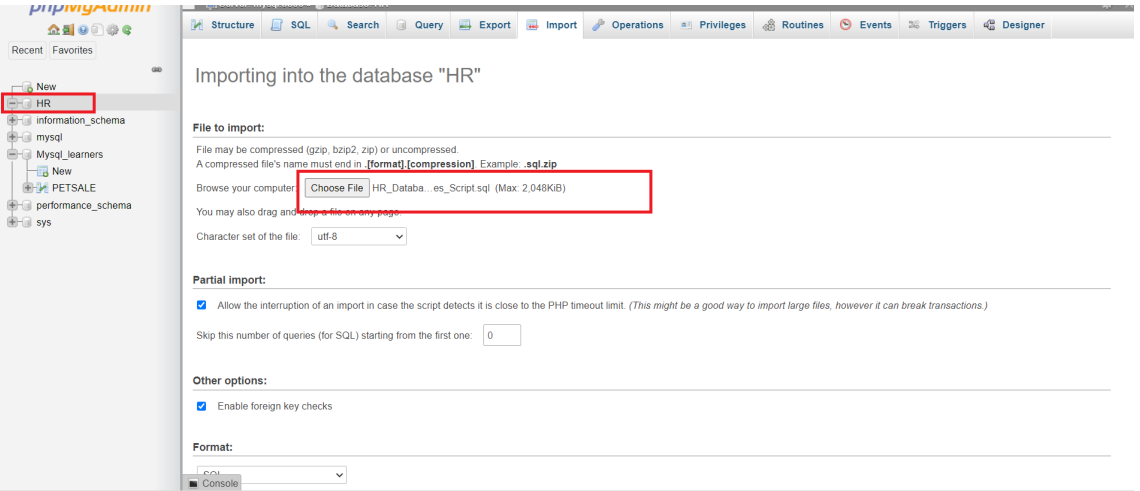
1. Click the link below to download the script file to your computer:

◦ [HR_Database_Create_Tables_Script.sql](#)

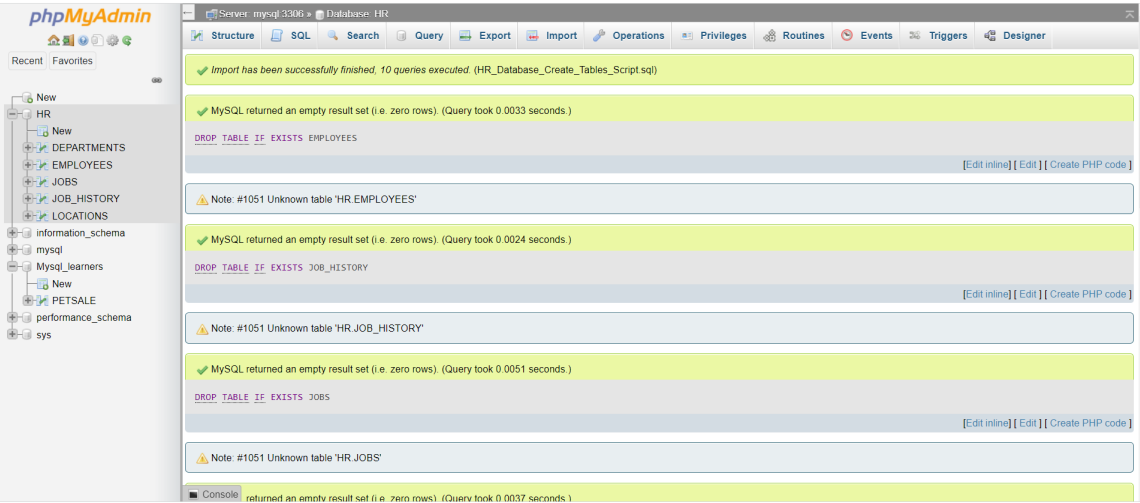
• Select the HR database. Later click on the Import tab.

• Click on **choose file**. Browse for the file and upload it.

• Later scroll down and click on **Go**.



• The script then gets imported successfully.



• Click on any of the tables and you will see its Table Definition (that is, its list of columns, data types, etc).

The screenshot shows the phpMyAdmin interface with the 'EMPLOYEES' table selected. The table structure is displayed in 'Table structure' view. The table has 11 columns: EMP_ID, F_NAME, L_NAME, SSN, B_DATE, SEX, ADDRESS, JOB_ID, SALARY, MANAGER_ID, and DEP_ID. The primary key is EMP_ID. The table is located in the HR database.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	EMP_ID	char(9)	utf8mb4_0900_ai_ci		No	None			Change Drop More
2	F_NAME	varchar(15)	utf8mb4_0900_ai_ci		No	None			Change Drop More
3	L_NAME	varchar(15)	utf8mb4_0900_ai_ci		No	None			Change Drop More
4	SSN	char(9)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
5	B_DATE	date			Yes	NULL			Change Drop More
6	SEX	char(1)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
7	ADDRESS	varchar(30)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
8	JOB_ID	char(9)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
9	SALARY	decimal(10,2)			Yes	NULL			Change Drop More
10	MANAGER_ID	char(9)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
11	DEP_ID	char(9)	utf8mb4_0900_ai_ci		No	None			Change Drop More

Exercise 2: Load data into tables

In this exercise, you will learn how data can be loaded into MySQL. You could manually insert each row into the table one by one, but that would take a long time. Instead, MySQL (and almost every other database) allows you to load data from .CSV files.

The steps below explain the process of loading data into the tables you created earlier in exercise 1.

1. Download the 5 .csv files below to your local computer:

- [Departments.csv](#)
- [Employees.csv](#)
- [Jobs.csv](#)
- [Locations.csv](#)
- [JobHistory.csv](#)

To load each table do the following steps.

- Select each table .
- Click on Import tab.
- Select the .csv file and click on Go to load the csv file.

The screenshot shows the phpMyAdmin interface with the 'EMPLOYEES' table selected. The 'Import' tab is active, and the 'Importing into the table "EMPLOYEES"' screen is displayed. The 'File to import:' section shows a file named 'Employees.csv' selected. The 'Format' dropdown is set to 'CSV'. The 'Other options' section has 'Enable foreign key checks' checked. The 'Format' dropdown is set to 'CSV'.

Importing into the table "EMPLOYEES"

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in **[format].[compression]** Example: **.sql.zip**

Browse your computer: Employees.csv (Max: 2,048KB)

You may also drag and drop a file on any page.

Character set of the file:

Partial import:

☒ Allow the interruption of an import in case the script detects it is close to the PHP timeout limit. (This might be a good way to import large files, however it can break transactions.)

Skip this number of queries (for SQL) starting from the first one:

Other options:

☒ Enable foreign key checks

Format:

Once the tables are loaded, you will get a message that the records are inserted successfully.

Further you can click on browse and view the data of each table.

Congratulations! You have completed this lab, and you are ready for the next topic.

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